

Speed Settings for Position Mode

1. **Velocity limits** - Software speed caps
2. **Tilt angle limits** - How much the quad can tilt forward
3. **Acceleration limits** - How fast it reaches max speed
4. **Thrust limits** - Motor power available

Key Parameters to Increase Speed

Parameter	Default	Default (km/h)	Range	Description
MPC_VEL_MANUAL	10 m/s	36 km/h	3-20 m/s	Max speed in Position mode
MPC_XY_VEL_MAX	12 m/s	43 km/h	0-20 m/s	Absolute max horizontal speed
MPC_MAN_TILT_MAX	35°	-	1-70°	Max tilt angle in manual modes
MPC_TILTMAX_AIR	45°	-	20-89°	Max tilt angle in flight
MPC_ACC_HOR_MAX	5 m/s ²	-	2-15 m/s ²	Max acceleration

Important relationships: - MPC_XY_VEL_MAX must always be greater than or equal to MPC_VEL_MANUAL - MPC_TILTMAX_AIR should be greater than MPC_MAN_TILT_MAX to give the flight controller reserve authority for wind compensation and corrections

Vertical Speed

Parameter	Default	Default (km/h)	Range
MPC_Z_VEL_MAX_UP	3 m/s	11 km/h	0.5-8 m/s
MPC_Z_VEL_MAX_DN	1.5 m/s	5.4 km/h	0.5-4 m/s

Position Mode Behavior

Default: MPC_POS_MODE = 4 (Acceleration based)

How to Change Parameters

Using QGroundControl:

1. Connect to drone
2. **Vehicle Setup** → **Parameters**
3. Search for parameter (e.g., MPC_VEL_MANUAL)
4. Change value → **Save**